Note Title
筆記標題

Your Name

October 16, 2026

Preface

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1. Introduction

Lecture 1: First Lecture

§1.1. Theorem Environments

13 Oct. 08:00

Definition 1.1.1 (Natural number)**.** The set of **natural numbers** is denoted $\mathbb{N}$.

Lemma 1.1.2 (Useful lemma)
Given the axioms of natural numbers $\mathbb{N}$, we have

$$0 \neq 1.$$

An obvious proof. Obvious. ■

Proposition 1.1.3 (Useful proposition)
From Lemma 1.1.2, we have

$$0 < 1.$$

Theorem 1.1.4 (Mass-energy equivalence)
Given Proposition 1.1.3, we then have

$$E = mc^2.$$

Proof. The blank left for me is too small, hence we put the proof in Appendix A.1.

Proof of key step. This inner proof uses ■ to distinguish from the outer ■. ■

■

Corollary 1.1.5 (Riemann hypothesis)
The real part of every nontrivial zero of the Riemann zeta function is $\frac{1}{2}$.

Proof. Left to the reader. ■ DIY

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§1.2. Other Environments

Exercise 1.2.1. Prove that $1 < 2$.

Solution. Iteratively apply Proposition 1.1.3. ■

Example 1.2.2 (Interesting example)**.** We note that $1 \neq 2$!

Note 1.2.3 (Important note)**.** As a consequence, $2 \neq 3$ also.

Remark 1.2.4. See that Proposition 1.1.3 is powerful. We also have a citation: [New26].

As previously seen . We saw Lemma 1.1.2 is helpful.

Claim 1.2.1. The center $Z(G)$ is always a normal subgroup of G.

Proof. For any $z \in Z(G)$ and $g \in G$, we have $gzg^{-1} = z \in Z(G)$. ■

Intuition . Key idea: think of the center as the fixed points of conjugation.

Internal links

You can use `\autoref` for internal references and `\cite` for bibliography entries.

§1.3. Footnote Test

Theorem 1.3.1

Every group of prime order is cyclic.[a]

[a] This is a footnote inside a theorem box.

Example 1.3.2. The integers $\mathbb{Z}$ form an infinite cyclic group.[1]

§1.4. Commutative Diagrams

Example 1.4.1. A commutative square drawn with `tikz-cd`:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle h} \\ C & \xrightarrow[k]{} & D \end{array}$$

§1.5. Vocab and Dividers

Use `\vocab{term}` to highlight a **key term** in blue bold.

Use `\hrulebar` to insert a visual section divider:

Content continues after the divider.

[1] This is a footnote inside an example box.

2. Chapter Title

Lecture 2: Second Lecture

§2.1. Section Title

20 Oct. 08:00

Content goes here.

§2.2. Source Marker Demo

Use `\source{}` to mark where each block of notes comes from.

slides p1

這裡寫對應第一頁簡報的筆記。可以用任何環境，例如：

Note 2.2.1. 這是第一頁的重點。

slides p2

這裡接著寫第二頁的筆記。內容可以連續延伸，`\source{}` 只是一個小標記。

book ch.3

也可以標記書本章節，不限於簡報。

3. Chapter Title

Lecture 3: Third Lecture

§3.1. Section Title

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Content goes here.

Appendix

A. Additional Proofs

§A.1. Proof of Theorem 1.1.4

We can now prove Theorem 1.1.4.

Proof of Theorem 1.1.4. See here. ■

Bibliography

[New26] I. Newton. *Philosophiae naturalis principia mathematica*. Innys, 1726. URL: https://books.google.com/books?id=WeZ09rjv-1kC.